

FUNCTIONAL SYSTEM DESIGN (FSD)

Tokopedia - Wishlist Feature

Document Version & Control

Field	Remarks
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Changelog

Version	Date	Changelog
v1.0	08/08/2026	Initial draft FSD for the Wishlist feature.
v1.1	30/09/2026	Aligned the FSD with the BRD after a requirement-by-requirement traceability check. Removed Wishlist sharing (out of scope in the BRD). Changed out-of-stock handling from auto-remove to an "Out of Stock" label. Made login mandatory. Added FRs for cross-device sync, removing items and quick add-to-cart. Rewrote NFRs with measurable targets. Fixed the data model and API contracts, and added a cache design and error handling.

Background / Problem Statement

Tokopedia does not have a Wishlist feature, so users can't save a product they like without adding it to the cart or buying it right away. The BRD shows that 34% of users leave a product detail page without taking any action, and more than 40% of them search for the same product again within 7 days. This FSD describes how the system will support the Wishlist requirements defined in the BRD.

From a technical standpoint:

- There is no table to store wishlist data.
- There is no endpoint to save, list or remove a wishlisted product.
- There is no way to notify users when a saved product changes price or goes on promotion.

Proposed technical solution:

- Add a new Wishlist Service with its own database.
- Cache each user's wishlisted product IDs in Redis, so the heart icon state loads quickly on search results.
- Listen to price and stock change events from the Product Service, and send notifications through the existing Notification Service.

Functional & Non-Functional Requirements

Functional Requirements

FR ID	FR Details	Priority	BRD Ref.
FR-01	The system must let a logged-in user save a product to their Wishlist by tapping the heart icon on the product detail page or in search results. Tapping it again removes the product.	High	Save to Wishlist button
FR-02	The system must show a Wishlist page listing all saved products, with the current price, the price when saved, and stock status.	High	Wishlist page
FR-03	The system must let users remove a product from the Wishlist page.	High	In-scope: remove products
FR-04	The system must keep Wishlist data in sync between the mobile app and web for logged-in users. A change on one device must show on the other within 5 seconds.	High	Cross-device synchronization
FR-05	The system must send a push and/or email notification when a Wishlist product drops in price or enters a promotional period.	Medium	Price/promo change notifications
FR-06	The system must show an "Out of Stock" label on Wishlist products that are unavailable. The product stays in the Wishlist so the user can see it once it's back in stock.	Medium	Stock status indicator
FR-07	The system must provide an "Add to Cart" button on each Wishlist item, without the user needing to open the product detail page.	Low	Quick add-to-cart
FR-08	If a guest user taps the heart icon, the system must ask them to log in, then save the product once login succeeds.	High	Supports FR-01 and FR-04

Non-Functional Requirements

NFR ID	Category	NFR Details
NFR-01	Performance	Add and remove requests must respond in under 300 ms at p95. Loading a Wishlist page of 20 items must take under 500 ms at p95.
NFR-02	Scalability	The service must support at least 10 million users with a Wishlist and handle 3x normal traffic during campaign peaks (e.g. 11.11 and 12.12) without breaching NFR-01.
NFR-03	Availability	The Wishlist Service must reach 99.9% monthly uptime. If the Notification Service is down, users must still be able to add, view and remove items.
NFR-04	Security	Every endpoint requires a valid bearer token. The user ID is taken from the token, never from the request body, so users can only read or change their own Wishlist. Data is encrypted in transit (TLS 1.2+) and at rest.
NFR-05	Notifications	A price or promo notification must be sent within 15 minutes of the change. A user gets at most one notification per product in 24 hours, to avoid spamming.

Out of Scope (per BRD)

- Sharing the Wishlist with other users or on social media.
- Organizing the Wishlist into custom folders or collections.
- AI-based product recommendations from Wishlist contents.
- Integration with Tokopedia's loyalty/points program.

System Overview

The Wishlist Service sits behind the API Gateway, which handles authentication. It stores data in its own database and uses Redis as a read cache. For notifications, it reacts to events from the Product Service instead of checking prices on a schedule.



Main flows

- **Add to Wishlist:** user taps the heart icon, the app calls API 1, the Wishlist Service saves the item with the current price, and updates the cache.
- **View Wishlist:** the app calls API 2. The Wishlist Service reads saved items and gets the current price and stock from the Product Service.
- **Price drop or promo:** the Product Service publishes a change event. The Wishlist Service finds every user who saved that product, checks the 24-hour notification limit, and asks the Notification Service to send the message.
- **Out of stock:** a stock event updates the item's status. The item stays in the Wishlist and shows the "Out of Stock" label.

Data Requirements

Table: `wishlist_item`

Field	Type	Description
wishlist_id	bigint	Primary key
user_id	bigint, not null	Logged-in user who saved the product (FK to users)
product_id	bigint, not null	Saved product (FK to products)
price_at_add	decimal(15,2)	Product price when it was saved, used to show and detect price drops
created_at	datetime	Time the product was added
last_notified_at	datetime, nullable	Last time the user was notified about this product (for the 24-hour limit in NFR-05)

Rules and relations

- One user can have many Wishlist items, and one product can appear in many users' Wishlists.
- Unique constraint on (user_id, product_id), so the same product can't be saved twice.
- Index on product_id, so the service can quickly find all users to notify when a product's price changes.
- Current price and stock are not stored here. They are read from the Product Service so they are always up to date.
- Only logged-in users can save items (FR-08). There is no guest Wishlist.

Cache (Redis)

Key	Value	TTL / Invalidation
wishlist:{user_id}	Set of product IDs the user has saved	24 hours; updated on every add or remove

The cache lets search results show a filled or empty heart for each product without querying the database for every item.

API Overview / Contract

All endpoints use the base path /api/v1 and require the header Authorization: Bearer <token>. The user is identified from the token.

API 1 - Add to Wishlist

Endpoint: POST /api/v1/wishlist

```
Request:
{
  "product_id": 98765
}

Response (201 Created):
{
  "status": "success",
  "wishlist_id": 1001,
  "message": "Product added to wishlist."
}
```

API 2 - Get Wishlist

Endpoint: GET /api/v1/wishlist?page=1&limit=20

```
Response (200 OK):
{
  "total": 2,
  "page": 1,
  "limit": 20,
  "wishlist": [
    {
      "wishlist_id": 1001,
      "product_id": 98765,
      "product_name": "Nike Air Running Shoes",
      "price_now": 850000.00,
      "price_at_add": 950000.00,
      "stock_status": "in_stock",
      "added_at": "2026-08-06T10:30:00Z"
    },
    {
      "wishlist_id": 1002,
      "product_id": 11230,
      "product_name": "Eiger 30L Backpack",
      "price_now": 450000.00,
      "price_at_add": 450000.00,
      "stock_status": "out_of_stock",
      "added_at": "2026-08-07T08:15:00Z"
    }
  ]
}
```

API 3 - Remove from Wishlist

Endpoint: DELETE /api/v1/wishlist/{wishlist_id}

```
Request:
DELETE /api/v1/wishlist/1001

Response (200 OK):
{
  "status": "deleted",
  "message": "Wishlist item 1001 has been removed."
}
```

Quick add-to-cart (FR-07)

No new endpoint is needed. The Wishlist page calls the existing Cart Service endpoint (POST /api/v1/cart) with the product_id, and the item stays in the Wishlist after it's added to the cart.

Error Handling

HTTP Code	When it happens	Message to user
401 Unauthorized	Missing or expired token	Please log in to use your Wishlist.
403 Forbidden	User tries to remove an item that isn't theirs	You can't change this item.
404 Not Found	Product or Wishlist item doesn't exist	This product is no longer available.
409 Conflict	Product is already in the user's Wishlist	This product is already in your Wishlist.
503 Service Unavailable	Wishlist Service is temporarily down	Something went wrong. Please try again in a moment.

Traceability: BRD to FSD

Each high-level requirement in the BRD is covered by at least one requirement or API in this FSD.

BRD High-Level Requirement	Priority	Covered by
Save to Wishlist button	High	FR-01, FR-08, API 1, Redis cache
Wishlist page	High	FR-02, FR-03, API 2, API 3
Cross-device synchronization	High	FR-04, NFR-01
Price/promo change notifications	Medium	FR-05, NFR-05, event flow in System Overview
Stock status indicator	Medium	FR-06, stock_status field in API 2
Quick add-to-cart from Wishlist	Low	FR-07, existing Cart Service endpoint

Open Questions

- Should there be a maximum number of items per Wishlist? A limit of 500 is suggested to protect performance; to confirm with the Product Manager.
- Can users choose between push and email notifications, or turn price alerts off? To confirm with the UI/UX Design Lead.
- What counts as a price drop worth notifying: any drop, or a minimum percentage? To confirm with the Product Manager.